01 The life science tools framework

FRAMEWORK

A product framework spanning academic research, biotech, pharma, CRDMOs and manufacturing.

Reagents

01	Recombinant proteins / antibodies
02	Enzymes / PCR and NGS reagents
03	DNA / RNA / chemical building blocks
04	Cell lines / culture media
05	Assay reagents for R&D and QC

Tests × dose per test × price

Consumables

01	Tips / microplates
02	Culture flasks / centrifuge tubes
03	Membranes / functional beads
04	Columns / chromatography resins
05	Gels / process components

Tests or batches × intensity × price

Instruments

01	Sequencers / liquid handlers
02	LC / LC-MS / mass spectrometers
03	Flow cytometers
04	High-content imaging systems
05	Cell-culture workstations

New installations + replacement + service

Use one product taxonomy; retain the scope of each market estimate.

Classify products separately from their use cases. Chromatography resins can be reused.

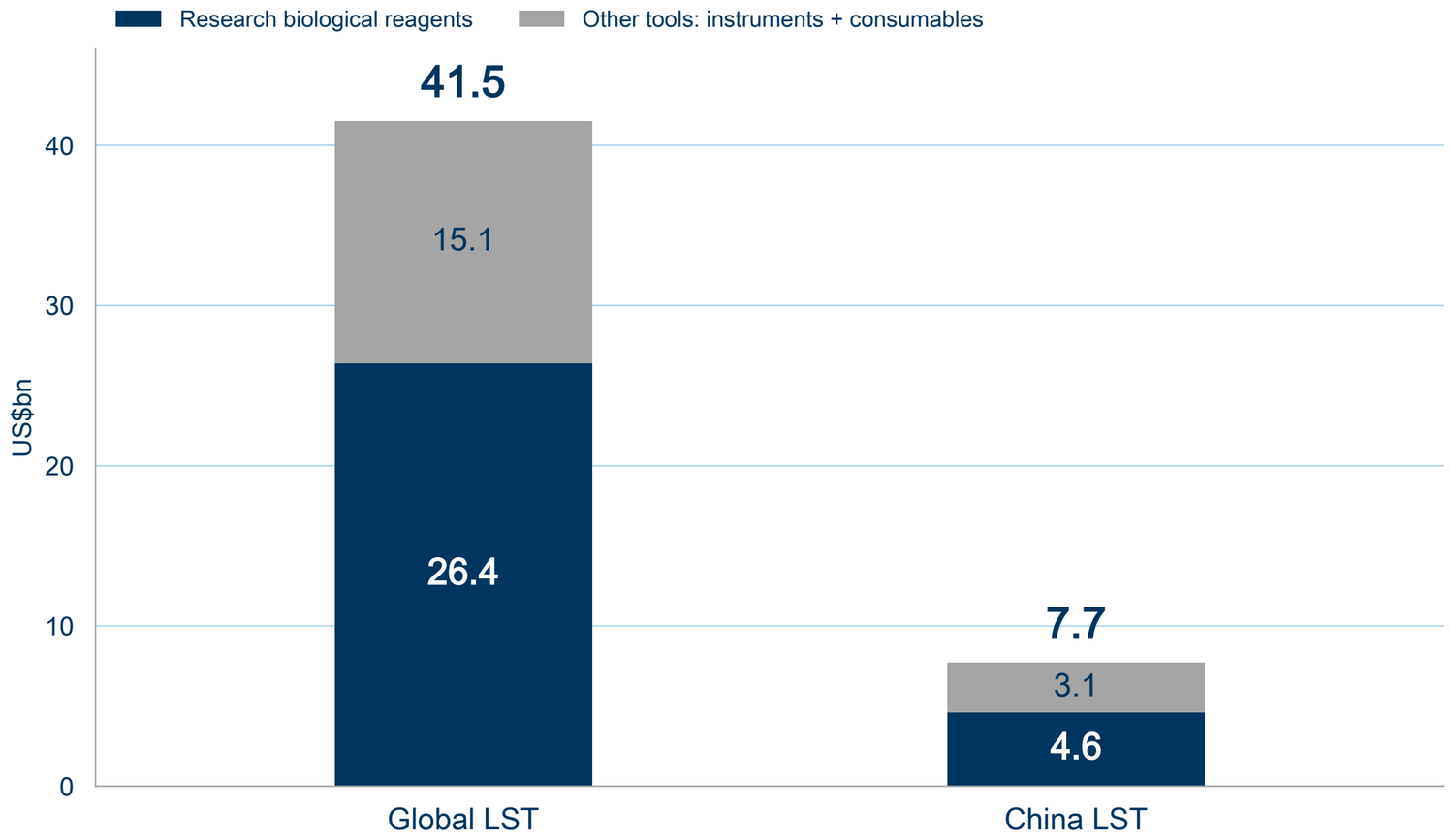
Source: supplied framework and prior product research. Existing TAMs do not automatically reconcile.

2026-10-12

02 2025 LST under one F&S definition

HISTORICAL

2025 F&S estimates commissioned for ACRO’s filing. A defined scope, not a broad-industry TAM.



China share within F&S scope

18.6%

7.7 / 41.5
Not vendor revenue exposure

Issuer-commissioned estimate.
Broader markets and vendor
China revenues use other scopes.

Other tools are the residual. Comparable instrument and consumable splits were not separately available.

Source: ACROBiosystems HKEX application, 30 Sep 2026, printed pp. 65 and 67; links in SOURCES.md.

03 Customer workflows across life science tools

USE MAP

A qualitative workflow map. Customer purchasing shares have not been established.

CUSTOMER	REAGENTS	CONSUMABLES	INSTRUMENTS
Academia / research hospitals	Targets, enzymes, cell media	Plates, tips, cultureware	Sequencing, imaging, LC-MS
Biotech / pharma R&D	Proteins, assay panels, expression reagents	Sensors, columns, filtration	BLI / SPR, automation, flow cytometry
CRO / CDMO	Discovery and process reagents	Labware, resins, scale-specific inputs	Assay, process and QC platforms
Biomanufacturing	Production media, feeds and enzymes	Resins, filters, single-use assemblies	Bioreactors, chromatography, QC instruments
IVD development	Antigens, antibodies, tool enzymes	Assay plates, magnetic beads	Research and QC instruments

An instrument supplier’s total China sales are not its China life-science sales.

Food, environmental and chemical-analysis uses excluded. Research instruments exclude clinical IVD analyzers.

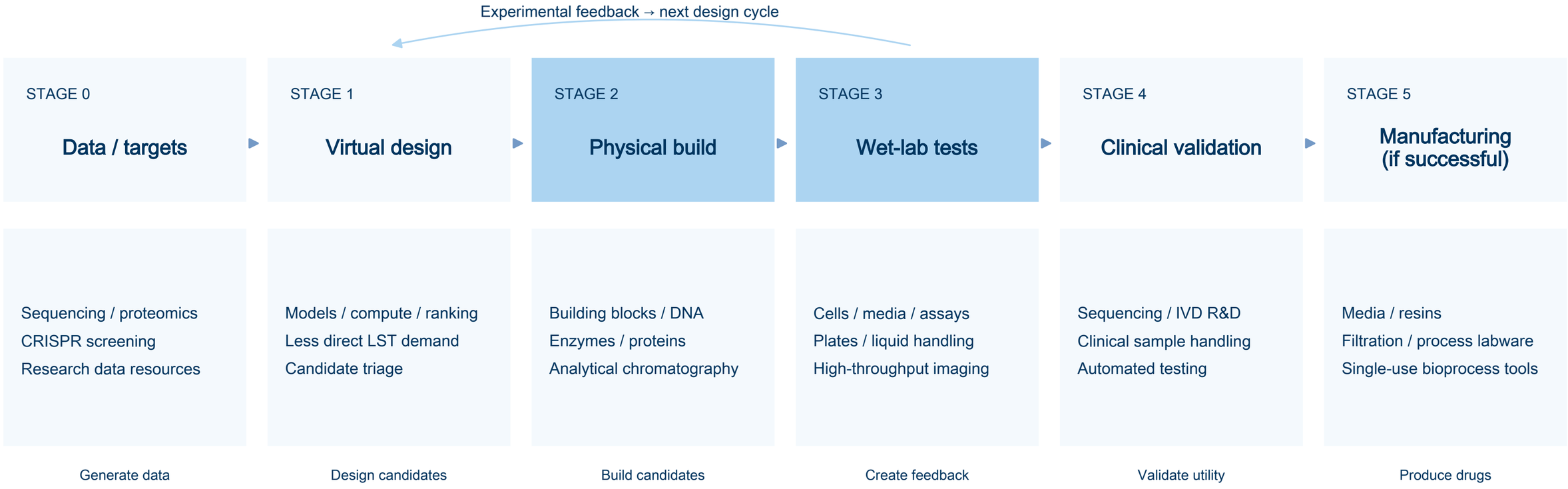
Source: representative supplier product uses; a role map, not an estimated customer revenue mix.

2026-10-13

04 AI discovery and its LST touchpoints

MECHANISM

AI candidates need physical construction and experimental feedback. Orders depend on funded projects.



The most direct potential uplift comes from building and testing candidates.

Stages 2–3 have more direct exposure. Clinical and manufacturing demand face attrition and time lags.

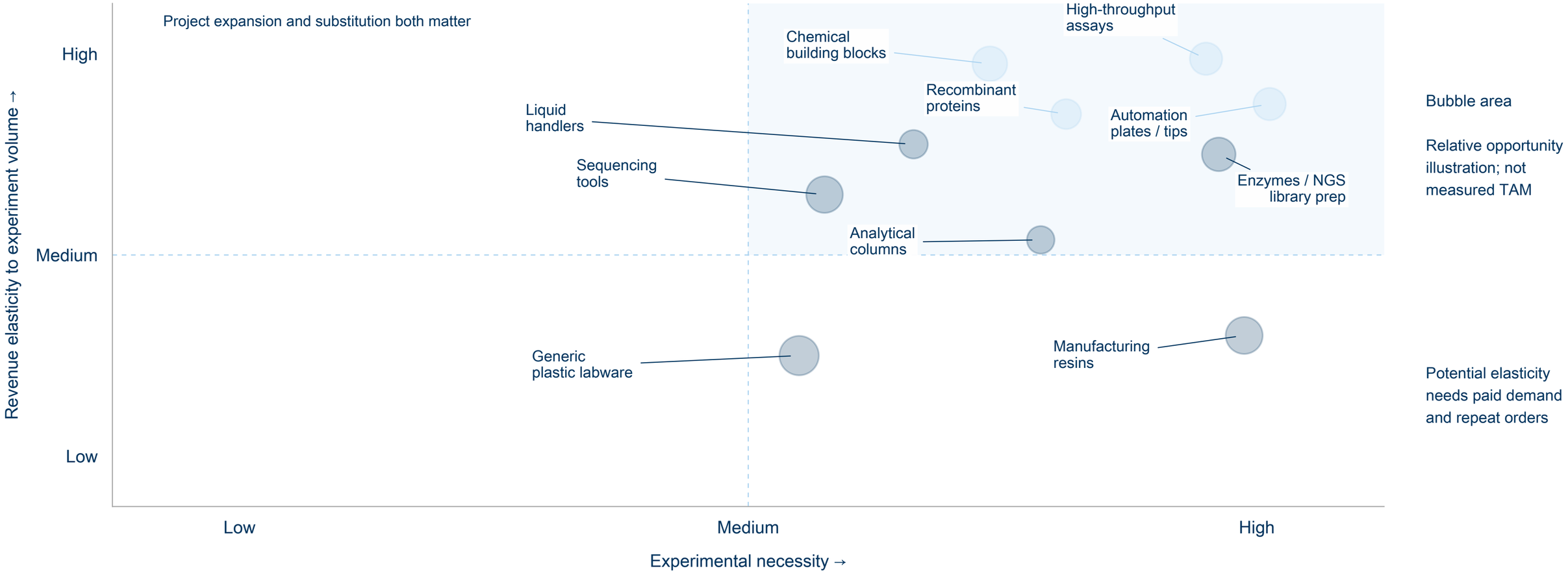
Source: supplied framework and product-use research. Arrows describe a mechanism, not realized orders.

2026-10-12

05 Which tools have more demand elasticity?

HYPOTHESIS

Positions and bubble areas are qualitative research judgments, not measured scores or additive TAMs.



Small-molecule synthesis can fall per project. Modality, assay depth, unit use and funded projects determine orders.

Source: supplied framework and analyst assumptions. Edit positions and relative areas in data.json.

2026-10-13

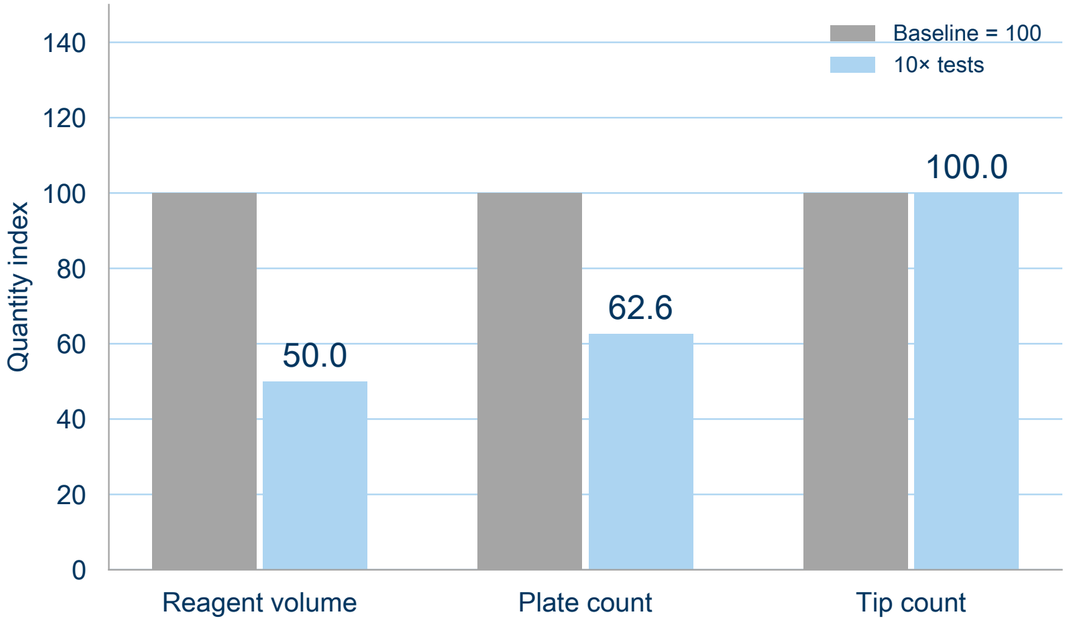
06 Quantities and prices in miniaturization

SENSITIVITY

10× tests: 10 L → 5 L reagent; 1,042 → 652 plates; 100k → 100k tips. Full usable plates.

Quantities × net unit prices determine spending

Physical inputs



Fewer plates can cost more

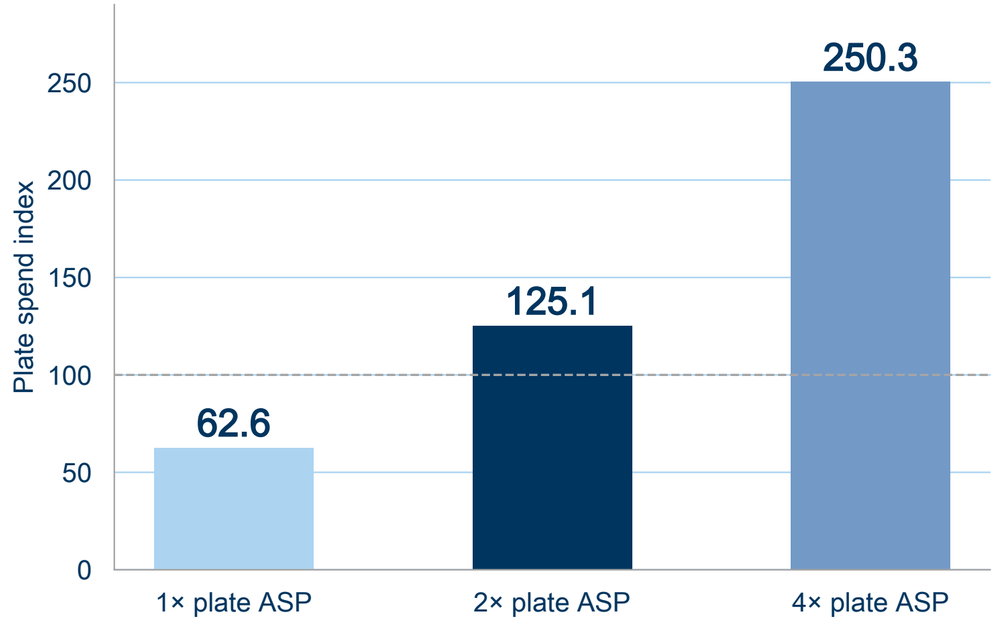


Plate spending exceeds baseline if the plate ASP ratio is above 1.60×.

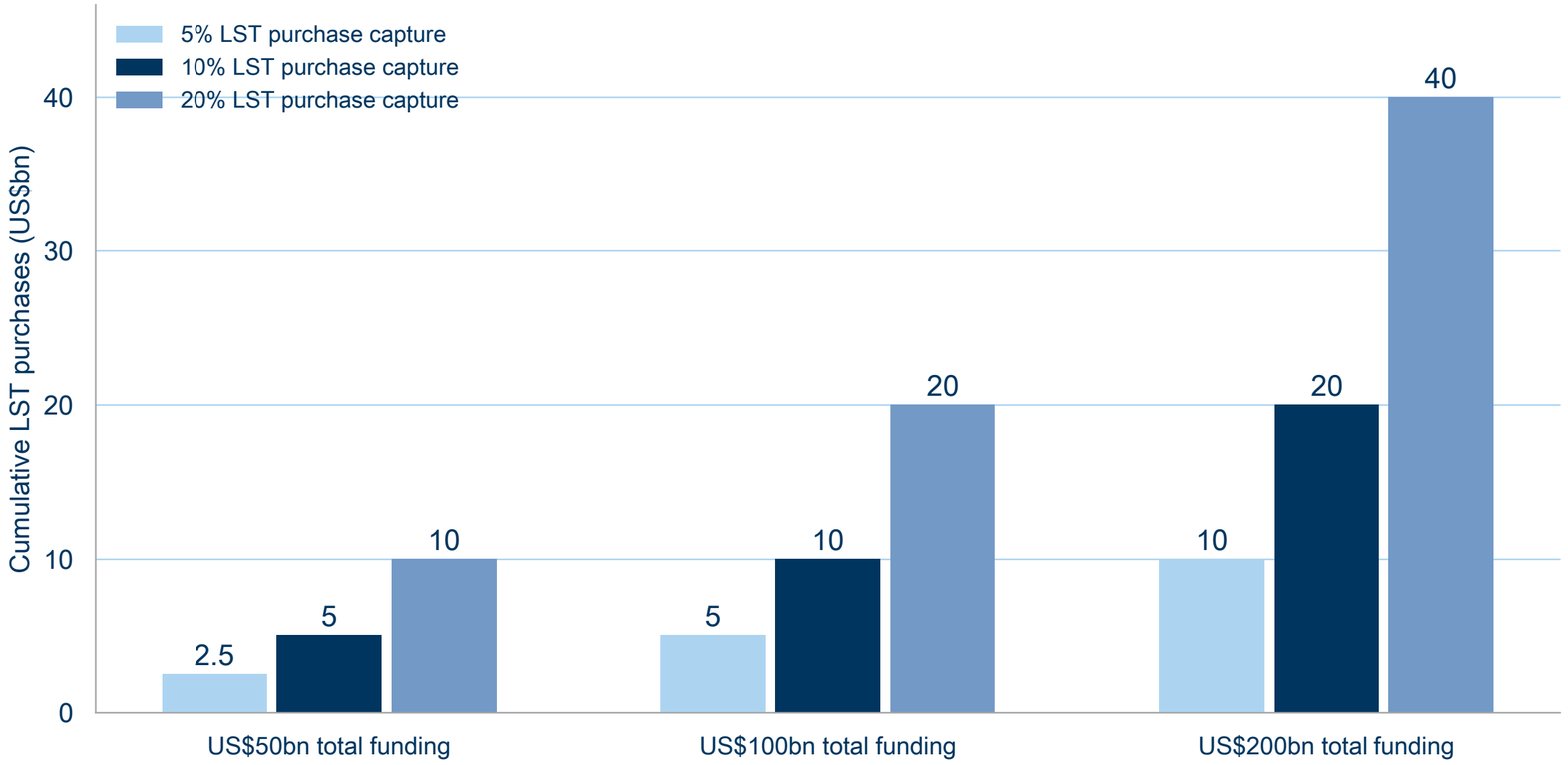
Price ratios are analyst inputs, not quoted ASPs. Dead volume, controls, usable wells and retesting can change quantities.

Source: transparent arithmetic from supplied test volumes and well counts. Units and sensitivities in SOURCES.md.

07 Funding-to-LST purchase sensitivity

SENSITIVITY

Hypothetical total funding pools. Only a fraction becomes purchases from LST suppliers.



Funding → tool purchases

R&D and production budgets exclude people, trials, software and unspent cash.

Capture includes tool purchases only. Count CRO/CDMO inputs once, separate from service fees.

Total capital raised is not an addressable tool market.

5% / 10% / 20% capture rates are model inputs. Cumulative scenarios, no fixed horizon; not observed funding or annual TAM.

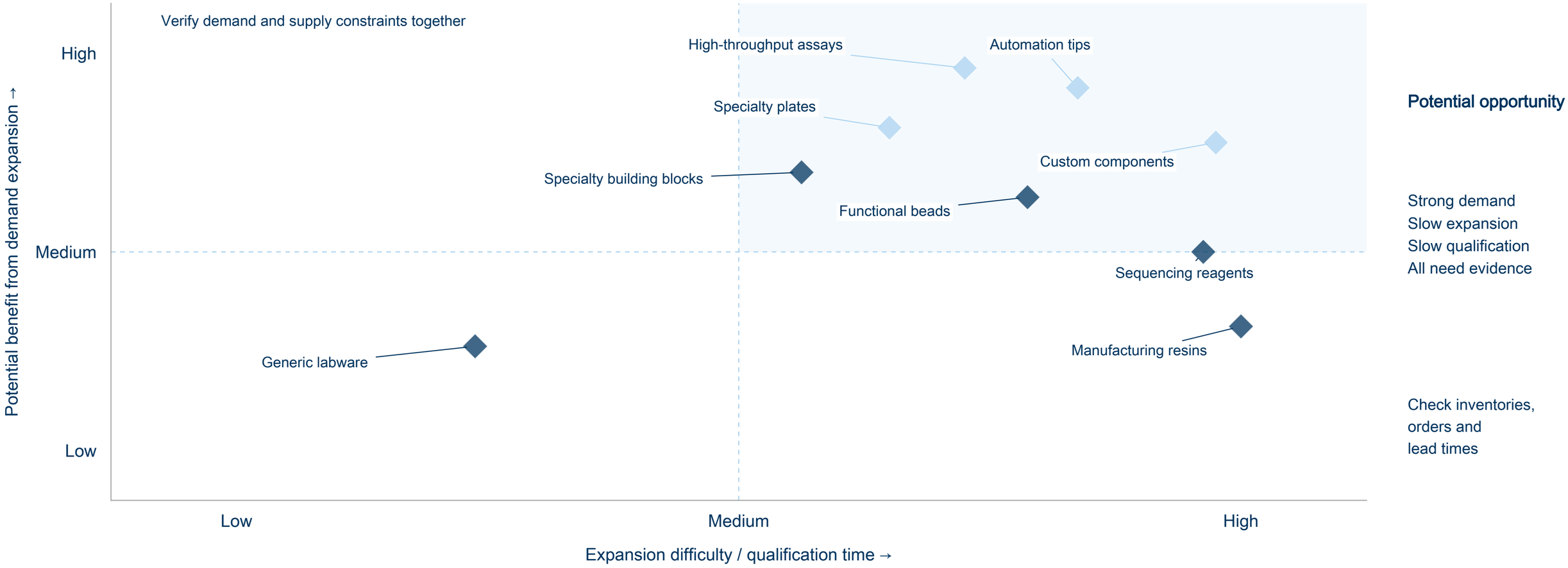
Source: analyst funding-to-purchase sensitivity. No empirical budget shares or equipment/reagent split is asserted.

2026-10-13

08 Potential supply constraints over 12–24 months

HYPOTHESIS

Relative expansion and qualification constraints over 12–24 months; current shortages are not established.



Track lead times, stockouts, qualification, yields and prices. Manufacturing resins have a longer demand lag.

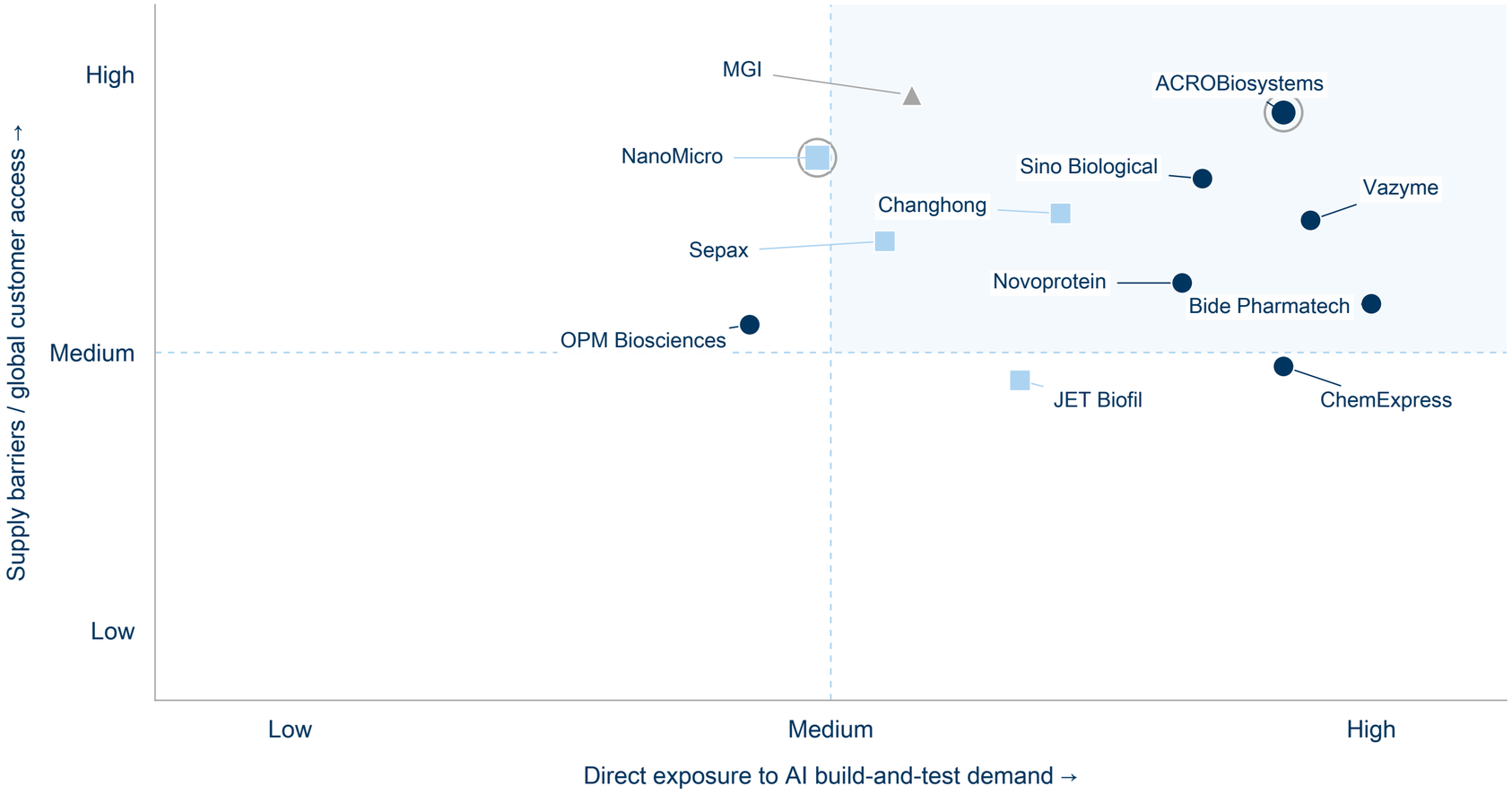
Source: supplied framework and analyst hypotheses. Technical barriers do not prove a capacity bottleneck.

2026-10-12

09 Mapping Chinese listed LST suppliers

HYPOTHESIS

Product exposure to AI build-and-test stages; positions do not measure AI revenue share or investment rank.



Reagents

Proteins, enzymes, blocks, media

Consumables

Labware, columns, resins

Instruments / platforms

Sequencing and automation

Two different demand pathways

ACRO: expression / functional assays
NanoMicro: purification + scale-up

Axes are qualitative judgments. Qualification, global customer access and paid orders still require evidence.

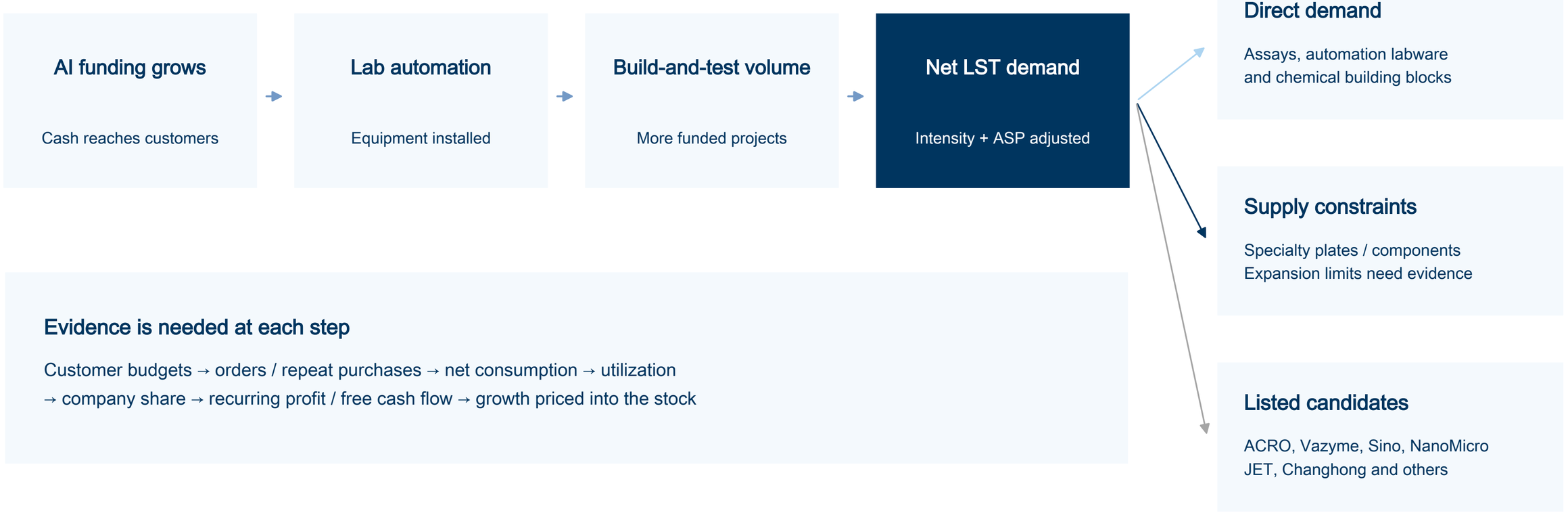
Sources: company filings and official product catalogs. Product rationale and assumptions are in data.json.

2026-10-12

10 From AI funding to LST shareholder returns

THESIS

Funding starts the chain. Projects, consumption, supply, market share and cash returns determine the outcome.



Evidence is needed at each step

Customer budgets → orders / repeat purchases → net consumption → utilization
→ company share → recurring profit / free cash flow → growth priced into the stock

Experimental relevance ∩ Evidenced supply constraints ∩ Global customer access

Seek experimental relevance, evidenced supply constraints and global access; test the valuation hurdle.

Source: supplied framework and prior company research. Capital announcements are not LST orders.

2026-10-12

AI discovery: workflows, evidence and sensitivities

Modalities use different units. Workload sensitivities and historical clinical benchmarks are shown separately.

STAGE	BEFORE GEN-AI	AI MECHANISMS	WHAT CHANGES
Virtual search / design	Large virtual libraries	Quality + selection	Large libraries predate generative AI. The investment mechanism is better selection and learning.
Physical samples	Program-specific	Modality-dependent	Small molecules can need fewer syntheses per program. Protein libraries and individual profiling are different units.
Assay workload	**1× baseline**	**1× / 2× / 5× / 10×**	**Analyst workload sensitivities, not predicted fully profiled candidate counts. Include an efficiency/no-growth case.**
Clinical development	Phase I programs	Funded clinical programs	Clinical biology, patients, capital and time remain constraints; discovery gains do not establish clinical gains.
Phase I to approval	7.9% / 13.8% (study estimates)	AI advantage unproven	BIO 2011–2020 and Wong 2000–2015 are different cohorts and methods, not an AI comparison.

Tool demand depends on funded programs, assay depth, net inputs and prices.

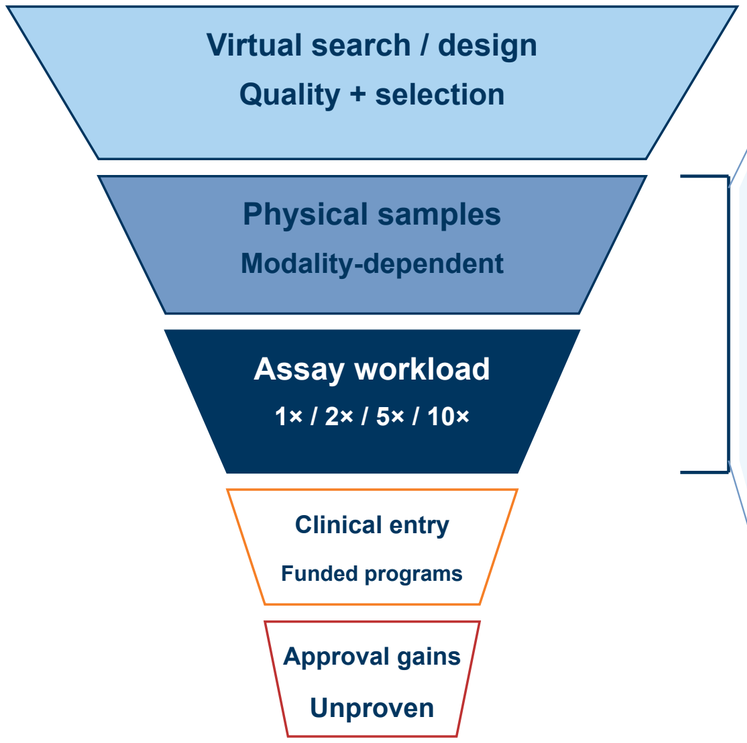
Sources: Lyu 2019; Sumitomo 2021; BIO 2011–2020; Wong 2000–2015. Workloads are analyst sensitivities, not empirical cohorts.

AI selection and biological evidence shape tool demand

Workflows vary by modality; paid demand depends on funded assays, depth, net inputs and prices.

MODALITY-AWARE DISCOVERY WORKFLOW

Schematic widths; workload multipliers are sensitivities



SMALL MOLECULES

Fewer syntheses per program are possible.

PROTEINS / ANTIBODIES

Pooled screens scale; individual profiling differs.

WET LAB = BUILD + TEST
No universal molecule-count funnel.
AI clinical advantage remains unproven.

The LST ecosystem behind the capacity gate

Issuer-cited subset estimates · different scopes / years · source conflicts disclosed below

Molecular construction

DNA / RNA synthesis, enzymes
Chemical building blocks

RMB 1.33bn

China • 2025 • DNA synthesis services [1]
Oligos + long fragments only

TWIST BIOSCIENCE · GenScript Make Research Easy

CHINA / HK LISTINGS	US LISTINGS / PARENTS
GenScript · 1548.HK	Twist · TWST
Vazyme · 688105.SH	Thermo Fisher · TMO
Tsingke · NEEQ:874813	

Protein & functional assays

Recombinant proteins, antibodies
Binding and activity assays

US$3.5bn

Global · 2025 · F&S RUO proteins [2]
Issuer-commissioned; estimate disputed

Acro BIOSYSTEMS

CHINA / HK LISTINGS	US LISTINGS / PARENTS
ACRO · 301080.SZ	Bio-Techne · TECH
Sino · 301047.SZ	Thermo Fisher · TMO

Cells & culture systems

Cell models, media, feeds
Bioreactors and cell expansion

Source pending

China • 2024 • bioprocess media [3]
Original filing unavailable; TAM withheld

Gibco / HyClone

CHINA / HK LISTINGS	US LISTINGS / PARENTS
OPM · 688293.SH	Gibco / Thermo · TMO
ACRO · 301080.SZ	Cytiva / HyClone · DHR

Labware & sample handling

Tips, plates, tubes, filtration
Automation-compatible labware

~RMB 23bn

China • 2024 • plastic labware [4]
Includes clinical and industrial uses

BIOFIL · BD

CHINA / HK LISTINGS	US LISTINGS / PARENTS
JET Biofil · 688026.SH	Corning · GLW
Changhong · 300151.SZ	BD · BDX

Measurement & analytics

Sequencing, LC / LC-MS, flow
Imaging and quality control

RMB 4.3bn

China • 2024 • sequencing tools [5]
Sequencing subset; 2024 policy baseline

MGI · BD flow → WAT†

CHINA / HK LISTINGS	US LISTINGS / PARENTS
MGI · 688114.SH	Illumina · ILMN
NanoMicro · 688690.SH	Agilent · A
	Waters · WAT†

Purification & scale-up

Analytical columns, process resins
Filtration and single-use systems

Source pending

China • 2024 • bioprocess resins [3]
Original filing unavailable; TAM withheld

NanoMicro · cytiva

CHINA / HK LISTINGS	US LISTINGS / PARENTS
NanoMicro · 688690.SH	Cytiva · DHR
Sepax · 688758.SH	Repligen · RGEN

Automation + data orchestration span every tool family — TECAN. SIX: TECN

† Ex-BD biosciences / flow moved to Waters on 9 Feb 2026; BD (BDX) here means specimen collection.
Protein TAM is issuer-commissioned / disputed. Media & resin TAMs withheld. Sources, scope and tickers: SOURCES.md.

Fewer molecules. Deeper experimental data.

AI may reduce candidate counts while increasing characterization per molecule

From molecule throughput to data density: a potential driver of life science tools demand

Existing data: staged screening

1,000 × 1 initial label

Illustrative early single-task dataset

Antibody	Binds?
Ab-001	Yes
Ab-002	No
Ab-003	Yes
Ab-004	Yes
Ab-005	No
Ab-006	Yes

Binding first; deeper tests for selected hits

Multi-assay characterization already exists

AI: richer, linked training data

300–1,000 antibodies · Up to 9 label types

Measure selected properties on shared molecules; fill the information gaps

Antibody	KD	kon	koff	expr	Tm	agg	poly	epi	cell
Ab-001	120	7200	1800	8400	5100	75	3200	9000	430
Ab-002	8100	640	5400	2200	9500	1600	4100	500	6900
Ab-003	2600	9200	410	6500	3100	7900	950	4800	8300
Ab-004	9800	3100	7400	1200	6800	380	5500	1900	9100
Ab-005	540	4700	8800	3600	2000	6200	10000	710	5700
Ab-006	6300	1400	3000	9900	4600	8700	180	7600	2400

Learn binding, expression, stability and functional relationships

Synthetic values: 1–10,000. Darker = larger; no biological units or quality ranking.

Reuse existing data

Normalize formats, conditions and access
Better training may need no new experiment

Stage 0: fill data gaps

Add missing assays and biological diversity
Create reusable model-training datasets

Active learning: measure selectively

Prioritize informative molecules and tests
Limit coverage of costly measurements

The nine labels: what each property measures

KD	Equilibrium binding strength	**kon**	How quickly binding forms	**koff**	How quickly the complex separates
expression	Antibody yield from a cell system	**Tm**	Thermal unfolding transition	**aggregation**	Molecules forming larger clusters
polyreactivity	Binding to multiple non-targets	**epitope**	The binding site on the target	**cell potency**	Functional activity in living cells

Illustrative counts. Nine label types are not nine independent assays; richer data need not come at unchanged cost.

Sources: Jain et al., 2017; Arsiwala et al., 2025; van Niekerk et al., 2026. Mechanism illustration; definitions and links in SOURCES.md.

Experimental data density and the LST ecosystem

Stage 0: biological data production creates a wet-lab use case

Learning developability requires comparable assay data and an integrated tool workflow

Stage 0: biological data production

Continuous input to training and iteration

Binding kinetics
KD / kon / koff

Expression & physicochemical properties
expression / Tm / aggregation

Specificity & cell function
polyreactivity / epitope / cell potency

Training records: sequence, target, conditions, raw curves, replicates, batches and failures

Train / improve → **Design / validate**

Assays inform models

Labels → assays → purchases

Tool revenue needs a bridge from label coverage to workflow volume and unit consumption

Construction & expression

DNA synthesis, vectors, expression, purification

Expression labels + samples for other assays

Count shared sample preparation once

Protein & functional testing

SPR / BLI, binding curves, epitope binning

Target proteins, sensors / chips, buffers

KD / kon / koff can share one kinetic run

Cells & culture systems

Functional cells, media, transfection, expansion

Cell potency; selected polyreactivity assays

Record cell model, target level and conditions

Measurement & analysis

SEC / HIC, nanoDSF, LC-MS, flow cytometry

Stability, aggregation, characterization / QC

Columns are reused; model instrument CAPEX separately

Sample handling, labware & automation

Plates, tips, tubes, filtration / beads, liquid handling

Across all workflows; miniaturization, reuse and scheduling alter net consumption

ACROBiosystems: target proteins, functional cells & characterization

AIDD-Ready offering exists; paid orders and repeat purchases still need verification

NanoMicro: analytical separation & lab purification materials

Track sample runs, column life and ASP; manufacturing resin demand has a longer lag

LST purchases ≈ Σ [workflow executions × net purchased inputs per run × ASP]

Instrument CAPEX is separate. Product alignment does not establish AI orders or revenue. Sources: Ginkgo, Sartorius and company websites.

Typical protein / antibody screening workflow

From DNA constructs to functional evidence: the life science tools behind each step

Example: sequence-defined antibody candidates screened against a protein target

BUILD · Make physical samples | **TEST · Find binders** | **AI learning needs depth per molecule** | **LEARN · Iterate**

Potential constraint: comparable, reproducible assay data

01 / BUILD

Construct design / DNA synthesis

Encode antibody candidates as DNA constructs

OUTPUT
Sequence + construct

REAGENTS
Oligos / gene fragments
PCR enzymes, vectors
Cloning reagents

INSTRUMENTS
Thermocycler
Spectrophotometer
Liquid handling workstation

CONSUMABLES
PCR / deep-well plates
Tubes, pipette tips

REPRESENTATIVE SUPPLIERS
TWIST BIOSCIENCE · GenScript Make Research Easy
Tsingke · Vazyme
Thermo Fisher

02 / BUILD

Protein expression & purification

Produce candidate antibodies and target-protein samples

OUTPUT
Purified samples + yield

REAGENTS
Cells, media, transfection
Affinity resin, buffers

INSTRUMENTS
Shaker / bioreactor*
ÄKTA / FPLC system
Centrifuge

CONSUMABLES
Flasks / culture bags*
Columns, filters, tubes

REPRESENTATIVE SUPPLIERS
cytiva
GenScript · ACROBiosystems
Sino Biological · Gibco
OPM · Duoning

03 / TEST

Primary binding screen

Identify binders and rank initial binding signals

OUTPUT
Hit / binding signal

REAGENTS
Recombinant target proteins
Antibody candidates
Detection reagents, buffers

INSTRUMENTS
ELISA / plate reader
BLI or SPR platform

CONSUMABLES
96 / 384-well plates
BLI sensors / SPR chips
Sealing films, sample tubes

REPRESENTATIVE SUPPLIERS
ACRO BIOSYSTEMS
Sino Biological · Bioss
Sartorius · Cytiva · Tecan
JET Biofil

04 / TEST

Multi-property characterization

Evaluate binding, specificity and developability risks

KD · kon · koff · expression†
Tm · aggregation
polyreactivity · epitope binning

REAGENTS
Reference proteins, dyes
Assay panels, buffers
Analytical reagents

INSTRUMENTS
SPR / BLI; SEC-HPLC
LC / LC-MS
nanoDSF / DLS

CONSUMABLES
Columns, capillaries
Vials, filters, assay plates
Sensors / chips

REPRESENTATIVE SUPPLIERS
NanoMicro
Waters · Agilent · Thermo Fisher
NanoTemper · Sepax
SPR / BLI: Cytiva · Sartorius

05 / TEST

Cell-function validation

Test activity in a relevant cellular model

LABEL
Cell potency

REAGENTS
Cell models, media, cytokines
Staining antibodies
Reporter-assay reagents

INSTRUMENTS
Flow cytometer
High-content imaging
Plate reader

CONSUMABLES
Cell-culture plates / flasks
Pipette tips, filters

REPRESENTATIVE SUPPLIERS
BIOFIL®
Gibco · OPM · Vazyme · Bio-Techne
Molecular Devices · Corning
Waters Biosciences (ex-BD)

06 / LEARN

Data integration & feedback

Link assay results to the next AI design

OUTPUT
Traceable training data

REAGENTS
No direct assay reagents

SYSTEMS / SOFTWARE
LIMS / ELN
Linked data platform
Lab orchestration

CONSUMABLES
No direct assay consumables

REPRESENTATIVE SUPPLIERS
TECAN.
Hamilton
Benchling · Genedata

Sequence + assay context + results → next design cycle

Automation + data orchestration span every step

Wet lab = build + test. AI may test fewer molecules while increasing characterization depth per molecule.

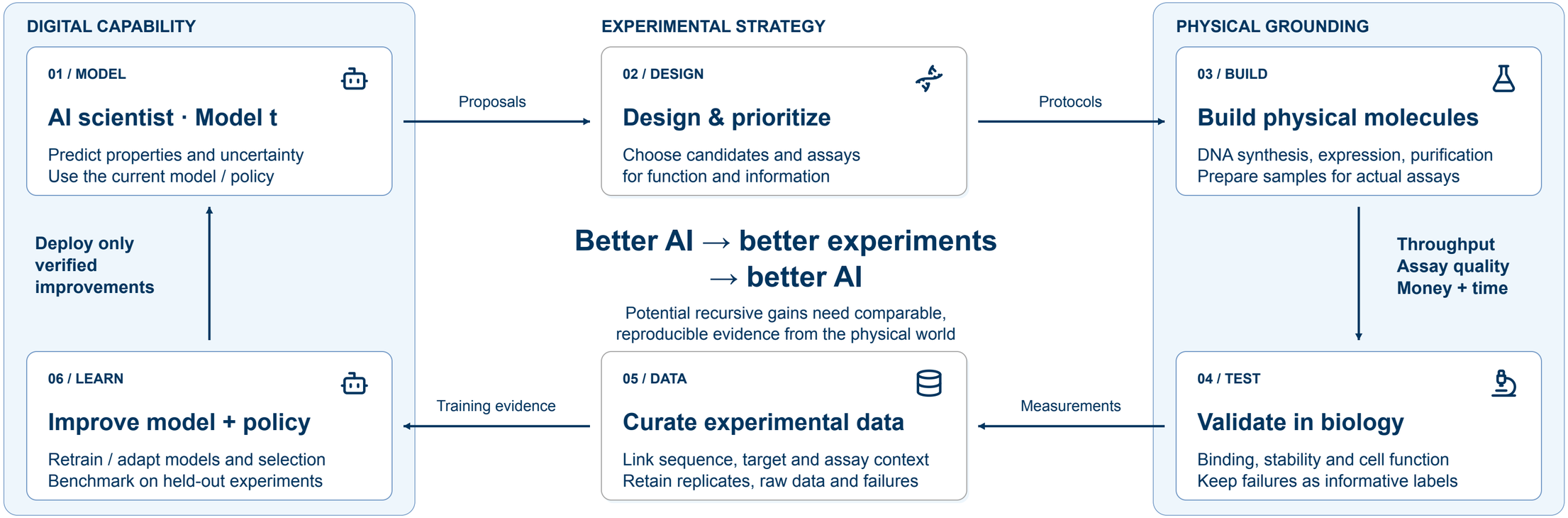
LST demand depends on molecule count × assay depth × net inputs per assay; instrument investment is separate.

Illustrative workflow; assays may run in parallel. * Scale-dependent. † Expression comes from stage 2. Supplier examples and primary sources: SOURCES.md.

RSI: AI design and biological validation

How experimental feedback could improve both biological designs and the AI design process

RECURSIVE SELF-IMPROVEMENT · CONCEPTUAL MECHANISM



LST supports every return to physical biology

BUILD
DNA synthesis · expression · purification

TEST
Proteins · biosensors · analytical columns
Cell assays · measurement platforms

CONNECT + SCALE
Labware · automation · data platforms
Comparable records across batches

The pace of improvement depends on useful experimental evidence per cycle

Potential LST demand ∝ funded cycles × assay depth × net inputs per assay

Automation and miniaturization can reduce inputs per measurement; instrument CAPEX is separate.

Conceptual RSI mechanism. Closed-loop learning does not establish sustained autonomous RSI. Primary sources: SOURCES.md.